

Infrastructure Requirements — Modernization Module

(Standalone On-Premises Deployment)

1. Minimum vs Recommended Hardware

Component	Minimum	Recommended
CPU	Intel Core i7 (8-core, 2.5 GHz+)	Intel Core i9 / Xeon (16-core, 3.0 GHz+)
RAM	32 GB DDR4	96 GB DDR4/DDR5
GPU	NVIDIA RTX 3060 (12 GB VRAM)	NVIDIA RTX 4070 SUPER (12 GB VRAM) or RTX 4090 (24 GB VRAM)
GPU VRAM	12 GB (required for qwen2.5-coder:14b)	24 GB (runs qwen3-coder:30b natively)
System Disk (OS + App)	200 GB SSD	500 GB NVMe SSD
Data / Model Disk	200 GB (Ollama models)	500 GB NVMe SSD (models + cache)
Network	100 Mbps (internal)	1 Gbps (for initial model downloads)

Why high RAM? Models like qwen3-coder:30b require VRAM + CPU RAM offloading. With 12 GB GPU VRAM, the remaining model layers load into system RAM. 96 GB ensures smooth offloading with headroom.

2. Operating System

Requirement	Specification
OS	Windows Server 2019 / 2022 (recommended) or Ubuntu 22.04 LTS
Architecture	x86-64 (64-bit) only
IIS	IIS 10.0+ with URL Rewrite Module 2.1 (Windows)
PowerShell	PowerShell 5.1+ (for watchdog service)
GPU Drivers	NVIDIA Driver 525+ (CUDA 12.x compatible)
CUDA Toolkit	CUDA 12.1+ (required by Ollama for GPU inference)
cuDNN	8.9+ (required for optimal inference throughput)

3. Software Runtime Stack

3.1 Python Backend

Package	Version	Purpose
Python	3.11 or 3.12	Runtime
FastAPI	0.111.0	REST API framework
Uvicorn	0.29.0	ASGI server
httpx	0.27.0	Async HTTP client (Ollama calls)
pymupdf	≥ 1.24.0	PDF document parsing
python-docx	≥ 1.1.0	Word document parsing
python-multipart	≥ 0.0.9	File upload handling
pyyaml	6.0.1	YAML config parsing
python-dotenv	1.0.1	Environment variable management

Estimated Python venv disk usage: ~150–250 MB

3.2 Node.js / Frontend Build

Component	Version	Purpose
Node.js	20 LTS (or 22 LTS)	Frontend build runtime
npm	10+	Package manager
Vite	5.4.11	Build bundler
React	18.3.1	UI framework

Frontend node_modules disk usage: ~800 MB–1 GB (only needed at build time)

Production build output (dist/): ~2–5 MB (served as static files)

3.3 Ollama (LLM Inference Engine)

Component	Specification
Ollama	v0.3.0 or later
Endpoint	http://localhost:11434
Protocol	HTTP REST + streaming
Model storage path	C:\Users\<user>\.ollama\models\ (Windows) or /usr/share/ollama/.ollama/models/ (Linux)

4. LLM Model Requirements

The Modernization module auto-selects the best available Ollama model in this priority order:

Priority	Model	VRAM Required	Disk Size	Quality
1 (Best)	qwen3-coder:30b	~17 GB (partial CPU offload)	~19 GB	Highest
2	qwen2.5-coder:14b	~8–10 GB	~9 GB	High
3	qwen2.5-coder:32b	~18 GB (partial CPU offload)	~20 GB	Highest
4	qwen2.5-coder:7b	~4–5 GB	~4.7 GB	Good
5	deepseek-coder-v2:16b	~9–10 GB	~9.1 GB	Good
6	codellama:34b	~20 GB	~19 GB	Moderate
7 (Fallback)	mistral:7b-instruct	~4 GB	~4.1 GB	Basic

Recommendation for On-Prem:

- **12 GB VRAM GPU:** Install `qwen2.5-coder:14b` as primary model.
- **24 GB VRAM GPU:** Install `qwen3-coder:30b` for highest quality output.
- Pull via: `ollama pull qwen2.5-coder:14b`

LLM inference parameters (configured in code):

Parameter	Value
Context window	16,384 tokens (adaptive per file)
Max output tokens	4,096 tokens
Temperature	0.10 (deterministic code generation)
Model keep-alive	10 minutes (stays hot in VRAM)
Request timeout	360 seconds per file
Parallel workers	5 (configurable via <code>MODERNIZATION_WORKERS</code>)

5. Disk Storage Breakdown

Component	Estimated Size	Notes
OS (Windows Server 2022)	~30 GB	Base OS
NVIDIA Drivers + CUDA	~5–8 GB	GPU compute stack
Ollama binary	~100 MB	Inference engine
LLM Models (1–2 models)	~10–40 GB	Stored in Ollama model dir
Python <code>.venv</code>	~250 MB	Module dependencies
Node <code>node_modules</code>	~1 GB	Build-time only
Frontend <code>dist/</code>	~5 MB	Served static files
Application code	~20 MB	Source + assets

Component	Estimated Size	Notes
LLM cache (%TEMP%)	~1–10 GB (grows)	Per-file conversion cache
Logs	~500 MB	Watchdog + uvicorn logs
Total (estimated)	~100–150 GB	Including OS, models, app

Recommended disk: 500 GB NVMe SSD to accommodate model updates, cache growth, and large project analysis.

6. Network Requirements

Requirement	Specification
Internal LAN	100 Mbps minimum (1 Gbps recommended)
Internet access (setup only)	Required for pip install, npm install, Ollama model downloads
Internet access (runtime)	NOT required — fully air-gapped after initial setup
Firewall — inbound	TCP 8084 (backend API), TCP 8090 (IIS/Nginx, if applicable)
Firewall — localhost	TCP 11434 (Ollama, internal only — must NOT be exposed externally)
Firewall — outbound	Block 11434 from external access

Security Note: Ollama listens on `localhost:11434` by default. Ensure firewall rules prevent external access to port 11434.

7. Environment Variables (Required)

Create a `.env` file in the `Modernization/` directory:

```
# Authentication (must match shared StratiQ secret if integrated, or set
standalone value)
AUTH_TOKEN_SECRET=<your-secure-random-string-min-32-chars>
AUTH_TOKEN_TTL_SECONDS=28800
AUTH_REQUIRED=true

# CORS – add the URL(s) where your frontend is served
CORS_ORIGINS=http://localhost:8090,http://<server-ip>:8090

# Optional: restrict file system browsing to a safe root
MODERNIZATION_FS_ROOT=C:\Projects

# Optional: tune parallel workers (default 5 each)
MODERNIZATION_WORKERS=5
MODERNIZATION_DOM_WORKERS=5

# Optional: domain cache TTL in seconds (default 86400 = 24h)
MODERNIZATION_DOM_CACHE_TTL=86400
```

```
# Backend bind settings
HOST=0.0.0.0
PORT=8084
```

8. IIS Configuration (Windows On-Prem)

Component	Requirement
IIS Version	10.0+
URL Rewrite Module	2.1+ (for <code>/api/mod/*</code> proxy rules)
ARR (Application Request Routing)	Required if using IIS as reverse proxy
Static file handler	For serving <code>frontend/dist/</code>
SPA fallback rule	Rewrite all non-API routes to <code>index.html</code>

URL Rewrite rule (in `web.config`):

```
<rule name="Modernization API Proxy" stopProcessing="true">
  <match url="^api/mod/(.*)" />
  <action type="Rewrite" url="http://localhost:8084/api/{R:1}" />
</rule>
```

9. Service Management

Component	Method
Backend startup	<code>watchdog_backend.ps1</code> (PowerShell watchdog)
Auto-restart on crash	Built into watchdog script
Run as service	Register via Windows Task Scheduler (at logon, highest privilege)
Ollama startup	<code>ollama serve</code> (register as Windows Service or Task Scheduler)
Log location	<code>Modernization\logs\</code> (watchdog + stderr)

Start command (manual):

```
# Start Ollama
Start-Process "ollama" -ArgumentList "serve" -NoNewWindow

# Start Modernization backend
.\Modernization\venv\Scripts\python.exe -m uvicorn api.server:app --host 0.0.0.0
--port 8084
```

10. Capacity Planning

Workload	Recommended Config
1–5 concurrent users, small projects (<50 files)	32 GB RAM, RTX 3060 (12 GB VRAM), i7 8-core
5–20 concurrent users, medium projects (50–500 files)	64 GB RAM, RTX 4070 SUPER (12 GB VRAM), i9 16-core
20+ concurrent users, large projects (500+ files)	96 GB RAM, RTX 4090 (24 GB VRAM), Xeon 16-core+
Enterprise / large monolith modernization	128 GB RAM, Dual GPU (A100/H100), multi-node

Throughput note: With `MODERNIZATION_WORKERS=5` and `qwen2.5-coder:14b`, expect ~2–10 seconds per file conversion. A 200-file project takes approximately 10–30 minutes end-to-end.

11. Pre-Deployment Checklist

- ☐ NVIDIA GPU driver (525+) installed and verified (`nvidia-smi`)
- ☐ CUDA 12.x toolkit installed
- ☐ Ollama installed and `ollama serve` running on port 11434
- ☐ Preferred LLM model pulled (`ollama pull qwen2.5-coder:14b`)
- ☐ Python 3.11+ installed and `pip` accessible
- ☐ Python venv created and dependencies installed (`pip install -r requirements.txt`)
- ☐ Node.js 20 LTS installed (for frontend build only)
- ☐ Frontend built (`cd frontend && npm install && npm run build`)
- ☐ `.env` file configured with secure `AUTH_TOKEN_SECRET`
- ☐ IIS URL Rewrite rules configured (or Nginx reverse proxy configured)
- ☐ Firewall: TCP 8084 open internally, TCP 11434 blocked externally
- ☐ Windows Task Scheduler entry created for watchdog at logon
- ☐ Disk storage: at least 150 GB free before setup

12. Summary — Minimum Viable On-Prem Deployment

Item	Specification
Server	1x physical or VM server
CPU	Intel i7 / Xeon, 8+ cores
RAM	32 GB DDR4 (96 GB recommended)
GPU	NVIDIA RTX 3060 12 GB VRAM (RTX 4070 SUPER recommended)
Storage	500 GB NVMe SSD
OS	Windows Server 2022 or Ubuntu 22.04 LTS
Runtime	Python 3.11, Node.js 20 LTS, Ollama, IIS 10 / Nginx

Item	Specification
LLM Model	qwen2.5-coder:14b (minimum), qwen3-coder:30b (recommended)
Internet	Required at setup; air-gapped at runtime
External services	None — fully self-contained